

13 Let the intensities of light after passing through X and Y be I_X and I_Y respectively.

By Malus Law,

$$I_Y = I_X \cos^2 \theta$$

$$I = I_Y \cos^2 \theta$$

$$= I_X \cos^4 \theta$$

Hence, the magnitude of I is the largest when θ is the smallest.

Answer: A